Thin smears thus prepared are air dried, fixed in alcohol and stained by one of the Romanowsky stains such as Leishman, Giemsa or JSB (Jaswant singh and Bhattacharjee) stain.

Thin smears are used for:

a. Detecting parasites, and

b. For determining the species of the infecting parasite.

Thick smear

They are prepared usually with 3 drops of blood spread over a small area of about 10mm. The thick film is dried. This smears consist of a thick layer of dehemoglobinized (lysed) red blood cells. It is not fixed in methanol.

Thick film is stained similar to thin film. Thick smears have the advantage of concentrating the parasites and therefore increase the sensitivity of diagnosis. Thick smears are used for:

a. Defecting parasites,

b. Quantitating parasitaemia, and

c. Demonstrating malarial pigments.

Fluoroscence microscopy

The method is mainly used for mass screening in field laboratory. Fluorescent dyes like acridine orange is used to stain the blood smears. It stains DNA as fluorescent green and cytoplasmic RNA as red.

QBC (Quantitative Buffy coat smear)

This is a sensitive method for detection of malaria parasites. In this method, blood is collected in a capillary tube coated with fluorescent dye and is subjected to centrifugation. After centrifugation, the Buffy coat in the centrifuged capillary tubes is examined under a fluorescent microscope. Acridine orange – stained malaria parasites appear brilliant green.

Serodiagnosis

It is not helpful in clinical diagnosis. It is used mainly for epidemiological survey and to identify the infected donors in transfusion malaria. The test used are indirect haemagglutination (IHA), Indirect fluorescent antibody (IFA) and Enzyme – linked immunosorbent assay (ELISA) for the defection of serum antibodies.

Rapid Antigen defection tests kits are available commercially like the dipstick, card and cassette bearing monoclonal antibody. These tests are based on the detection of antigens using immune chromatographic methods. These tests can detect plasmodium in 15 minutes.

Molecular diagnosis

DNA probe and PCR are highly sensitive methods for the diagnosis of malaria. It is more sensitive than that of thick blood smear. It is highly species specific.

Other tests includes the measurement of hemoglobin, total WBC and platelet count in severe falciparum malaria, urine can be tested for free hemoglobin, if black water fever is suspected. Blood urea and serum creatinine has to be monitored for renal failure.

8.5.14 Treatment

The most commonly used drugs are Chloroquine, Quinine, Pyrimethamine and Doxycycline.

8.5.15 Prevention and Control

The preventive measures to control malaria mainly depend on treatment of

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